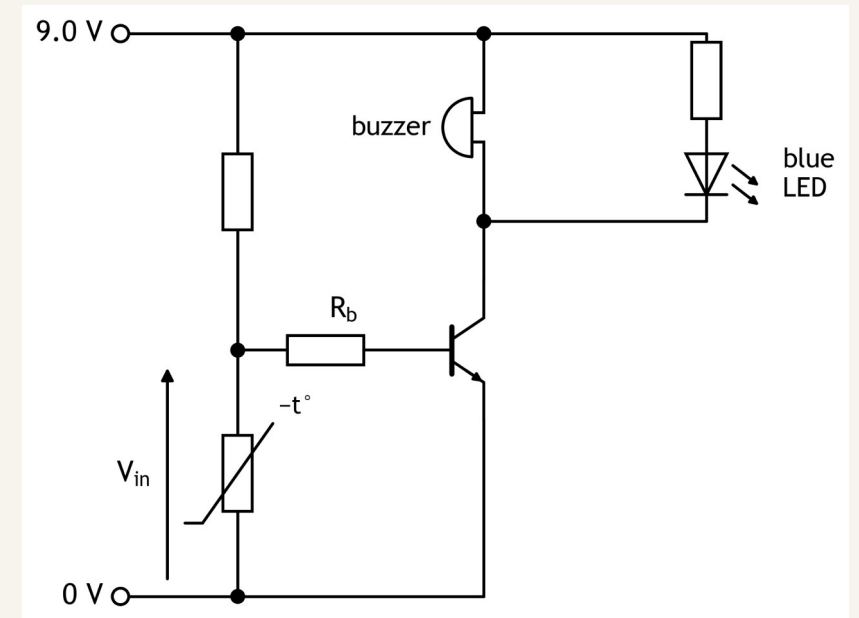


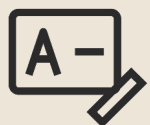
Checkpoint Revision

Analogue electronics and energy:
quick-fire practice.

National 5 Engineering Science



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

By the end of this starter...

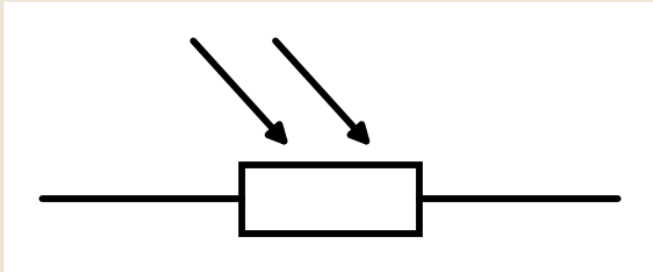


- 1 I can calculate resistance, current and voltage.
- 2 I can describe how a sensor and voltage divider switch a circuit.
- 3 I can explain transistors, relays, diodes and system boundaries.
- 4 I can calculate power, efficiency and heat energy.
- 5 I can describe engineers' roles, impacts and new technology.

Name each component. Give its full name.

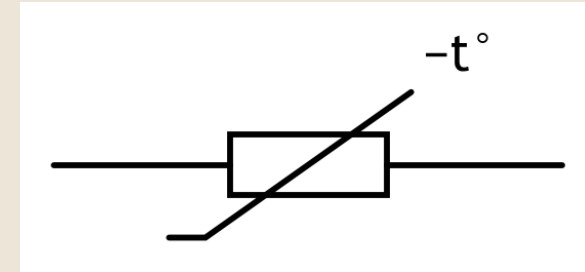


1



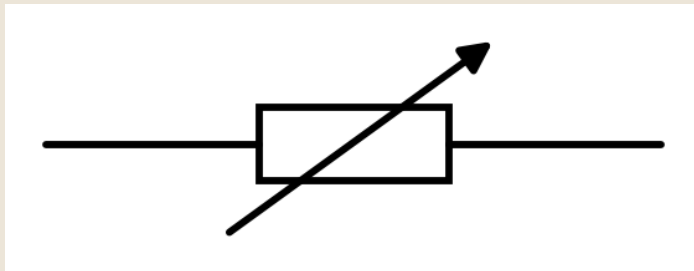
Light dependent resistor (LDR)

2



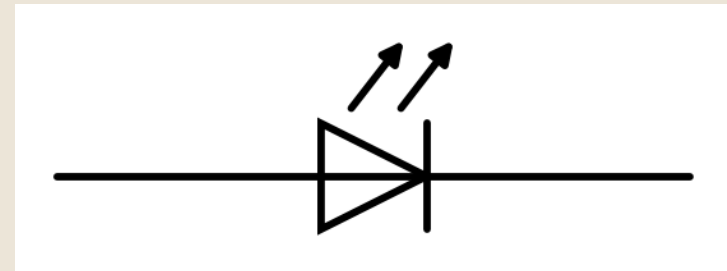
Thermistor

3



Variable resistor

4



Light emitting diode

Meters, resistance and Ohm's law



1

An ammeter is connected in ... with the component.

series

2

A voltmeter is connected in ... with the component.

parallel (across it)

3

Calculate R_t for $10\ \Omega$ and $22\ \Omega$ in series.

$32\ \Omega$

Meters, resistance and Ohm's law



4 Calculate R_t for $40\ \Omega$ and $40\ \Omega$ in parallel.

20 Ω

5 Calculate V when $I = 0.50\text{ A}$ and $R = 18\ \Omega$.

9.0 V

Sensors and dividers



1 Light level goes up. LDR resistance goes ...

down

2 Temperature goes up. NTC thermistor resistance goes ...

down

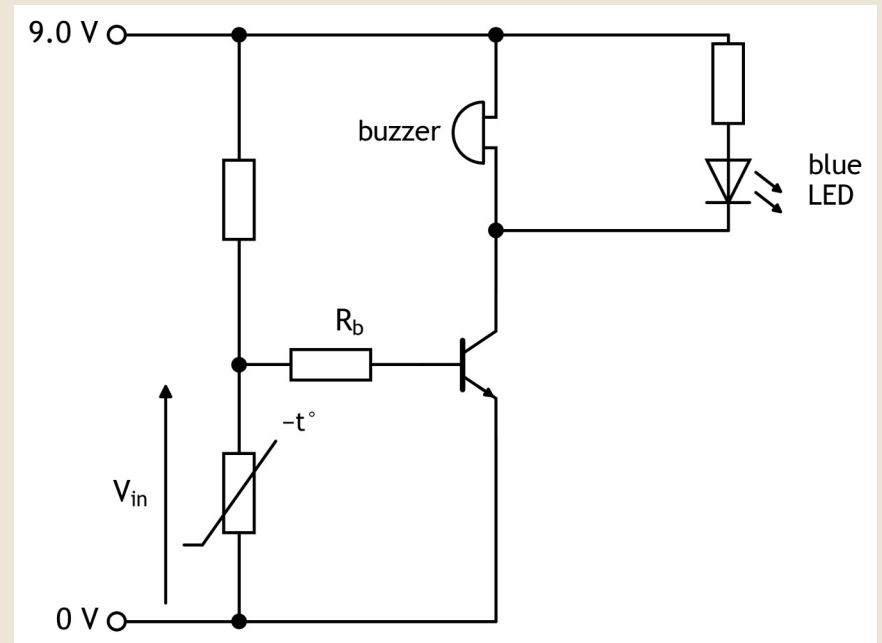
3 State the voltage across each of two equal resistors on a 10 V supply.

5.0 V

Describe the operation: frost alarm

Describe the operation as the temperature decreases.

1. The resistance of the thermistor increases.
2. V_{in} increases.
3. The transistor switches on.
4. The buzzer sounds.
5. The blue LED lights.



Transistor, relay, diode



1

A transistor switches on when V_{be} reaches about ...

0.7 V

2

Name the three connections of a transistor.

base, collector, emitter

3

A relay lets a small current switch a ... current.

large (larger)

Systems



- 1 A buzzer is an ... sub-system.
- 2 A microcontroller is the ... sub-system.
- 3 The dashed line round the sub-systems is the ...

output

process

system boundary

Power and energy



1 Calculate P when $V = 230 \text{ V}$ and $I = 2.0 \text{ A}$.

460 W

2 Calculate E when $P = 50 \text{ W}$ for $t = 60 \text{ s}$.

3000 J

3 The unit of energy is the ...

joule (J)

Efficiency and heat



1

Calculate η when $E_{\text{out}} = 300 \text{ J}$ and $E_{\text{in}} = 400 \text{ J}$.

0.75 (75 %)

2

Calculate E_h to heat 1.0 kg of water by 10°C .

41 800 J

3

State what ΔT means.

change (rise) in
temperature

Contexts



1 An emerging technology is ...

still being developed,
not yet widely used

2 Name two types of impact other than economic.

social and
environmental

3 State one advantage of simulating a circuit first.

faults found before
buying parts

How calculation marks are earned

Calculate the current. The supply is 3.0 V and the resistance is 15 Ω .

Relationship $V = IR$ 0 marks on its own

Substitute $3.0 = I \times 15$ 1 mark

Rearrange $I = 3.0 \div 15$ 1 mark

Answer + unit $I = 0.20 \text{ A}$ 1 mark

How describe and explain marks are earned

Explain = cause + effect

The resistor limits the current (cause) ... so the LED is not damaged (effect).

Describe the operation

One mark per link, in order: sensor, V_{in} , transistor, each output.

Scores 0

'Adjust VR' with no direction. 'LDR' when a full name is asked for.

A $40\ \Omega$ resistor and a $10\ \Omega$ resistor are connected in parallel. What is the total resistance?



- A $50\ \Omega$
- B $25\ \Omega$
- C $0.125\ \Omega$
- D $8.0\ \Omega$

Answer D: $(40 \times 10) / (40 + 10) = 400 / 50 = 8.0\ \Omega$. It is less than $10\ \Omega$, as it should be.

An LDR is the bottom resistor of a voltage divider. V_{in} is measured across the LDR. It gets darker. What happens?



- A LDR resistance decreases; V_{in} decreases
- B LDR resistance increases; V_{in} decreases
- C LDR resistance increases; V_{in} increases
- D LDR resistance decreases; V_{in} increases

Answer C: In the dark the LDR's resistance increases. It takes a bigger share of the supply, so V_{in} increases.

How must the protection diode be connected?



- A In series with the relay coil
- B Across the coil, forward biased
- C Across the base resistor
- D Across the coil, reverse biased

Answer D: Across the coil, reverse biased: the bar (cathode) points to the positive supply.

A motor is supplied with 40 000 J. It is 0.85 (85 %) efficient. What is the useful output energy?



- A 34 000 J
- B 47 059 J
- C 6 000 J
- D 3 400 000 J

Answer A: $E_{\text{out}} = \eta \times E_{\text{in}} = 0.85 \times 40\,000 = 34\,000 \text{ J}.$

Look back — by the end of this starter...



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Course specification – 1 of 6

Analogue electronic control systems

- function and purpose within a circuit of: battery, switch, resistor, variable resistor, LDR, thermistor, LED, buzzer, diode, motor, lamp, ammeter and voltmeter
- describing the function of a circuit in terms of input, process and output
- calculations involving the relationship between voltage, current and resistance (Ohm's Law)

Course specification – 2 of 6

Analogue electronic control systems

- calculations involving resistors in series and parallel
- calculations of voltage, current, and unknown values in a fixed-voltage divider
- designing a voltage divider to provide an input signal for a control circuit
- interpreting information given of characteristics for an LDR and an NTC thermistor

Course specification – 3 of 6

Analogue electronic control systems

- function of relays
- function of a protection diode in an electronic circuit
- explaining the switching function of a transistor
- operating an electronic control circuit, which includes a variable voltage divider, transistor, relay and output transducer

Course specification – 4 of 6

The systems approach

- systems and sub-system diagrams
- interaction of sub-systems

Energy and efficiency

- calculations involving forms of energy (kinetic, potential, electrical and heat)

This starter uses electrical and heat energy only.

Course specification – 5 of 6

Energy and efficiency

- applied calculations involving efficiency, work done and power using:
 - $E_w = Fd$ $P = E/t$, $E_k = \frac{1}{2} mv^2$ $E_p = mgh$
 - $E_e = VIt$ $E_h = cm\Delta T$ Efficiency $\eta = E_{out}/E_{in} = P_{out}/P_{in}$

Calculations

- manipulating given formulae to obtain answers

Course specification — 6 of 6

Engineering roles and disciplines

- varied roles of engineers in designing, implementing, testing and controlling complex systems

Impacts of engineering

- examples of social and economic impacts (positive and negative) of engineering
- explaining how emerging technologies may provide improved solutions to engineering challenges

Credits

Diagrams: Circuit and system diagrams drawn by R Stewart.

Course specification: © Qualifications Scotland (SQA). Course specification v2.0, pp.4-5